PROG8431 – Data Analysis, Mathematics, Modeling and Algorithms

Problem Analysis Workshop B

K-Means Clustering

DUE BEFORE NOV 5; 10PM

1. Instructions

1. Read and understand Unit 4 (Time Series) and Unit 5 (Clustering).
2. Make sure you can run the Clustering demonstration code (on the website under Week 7)
3. As a team, collaboratively write a Jupyter Notebook that:
    a. Starts with a markup cell with your team member names and ID numbers.
    b. Contains all the setup and configuration steps necessary to run the code that follows.
    c. Refine the data source from the previous assignments, and, if necessary, find others to add, clean up, or merge with.
    d. Follow the patten in the Clustering demonstration code, this time applied to your term project.
    e. In a markdown cell, explain how you will use clustering to deepen your understanding of the data from term project.
    f. Choose two variables to define your clusters (these will be the 'centres). In a markdown cell, explain your reasoning.
    g. Examine, describe and then standardize your data using one of the two methods discussed in class and provided in the demonstration code. In markdown cell, explain your reasoning.
    h. Run the k-means algorithm demonstrated in class and use an elbow chart as well as other assessments to determine the final number of clusters. In a markdown cell, explain your reasoning.
    i. Create a summary of the clusters, give them appropriate descriptive names and discuss what insights they have given you about your data in a markdown cell.
    j. **NOTE – There are many algorithms available in Python to conduct k-means clustering and to evaluate the results. If you use algorthims other than the ones demonstrated and explained in class, you will be expected to answer questions fully explaining the differences between your choice and the class exercises, the benefits and the details and your rationale for using different tools.**
4. Prepare a seven minute presentation of the work done in these steps.

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a. There is no need to prepare PowerPoint slides; you will present using the Jupyter Notebook.
b. Ensure all team members can connect their laptops to the classroom projector and be ready to present for the team if asked to.
c. Everyone in the team must know how the code was written, should understand it, and must be able to make changes to it.

2. Submission Guidelines

Submit the Jupyter Notebook in to the appropriate assignment folder. **Do not zip the file.** Also, include the link to the GitHub repository.
You may make as many submissions as you like, but only the final submission will be graded.

3. Presentation Guidelines

Use the same Jupyter Notebook you submitted to do your presentation. Penalties will apply if you use a different one.

Your presentation will be done in the time slot assignment by your instructor. The 7 minute timing is mandatory. The instructor will start a timer that will ring at the 7-minute mark.

All team members should be ready to answer questions about the materials and code you presented.

Your mark will be based on the accuracy of your code, the clarity of your data exploration, your ability to answer questions and your presentation within the 5-minute time slot.

Remember the discussion forums on eConestoga are a great place to ask questions.

4. Quiz

Following the presentations there will be a brief, individual quiz using the Lockdown browser and covering Clustering and Time Series (Weeks 6 and 7) as discussed in class.

5. Grading

This assignment will be marked based on the rubric below and is worth 10% of your total grade in the course.

Any team member not showing up on time for the presentation will not receive credit for the assignment.

A Jupyter Notebook submitted after the deadline but prior to the presentation will receive a 20% reduction in grades.

RUBRIC

Assessment	Item	Marks

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Group	Jupyter Notebook was correctly developed as a guide for a smooth presentation.	2
Group	Data Transformation - Data was transformed and standardized using an appropriate technique and justification.	4
Group	"Devlop a final recommended Clustering Scheme with full justification for: 1. Variables used to define the clusters 2. The number of clusters (i.e. the 'elbow' chart and scatter plots)	6
Group	Summary of Clusters using all data and insights are gained from them.	3
Group	Presentation: attendance, participation, and timing	2
Individual	Ability to answer questions	2
Individual	Collaboration: Team has a central hub for communications which is regularly used throughout the project life (e.g. MS Teams, GIT, etc.). This will be assessed individually based on contributions.	6
Individual	Quiz - Test on Time Series/Clustering	5

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